

#	Author(s)	Paper Title, Journal, & Year	Summary / Methodology	Strengths / Advantages	Weaknesses / Limitations	Research Gap / Future Scope
1	R. Kooi et al.	Large scale deep learning for computer aided detection of mammographic lesions, <i>IEEE Trans. Med. Imaging</i> , vol. 36, no. 4, pp. 1–10, 2020.	Presents a deep learning-based CAD system using CNNs trained on large datasets to identify suspicious regions in mammograms and improve accuracy.	High detection accuracy; automated lesion detection reduces manual effort; improves screening efficiency for radiologists.	Requires large annotated datasets; high computational cost/time; limited interpretability and data dependence.	Need for lightweight and explainable models with better clinical applicability.
2	H. Li, K. Wang, and X. Liu	Breast mass classification in mammograms using deep convolutional neural networks, <i>IEEE Access</i> , vol. 8, pp. 1–10, 2020.	A CNN-based method for classifying breast masses as benign or malignant by automatically extracting features from mammographic images.	High classification accuracy; enables automatic feature extraction; improves efficiency in mammogram analysis.	High computational complexity; requires large annotated datasets; limited generalization across different datasets.	Need for lightweight and generalizable models with improved interpretability.
3	Y. Wu, Y. Zhang, and J. Wang	Mammogram classification using multi-view convolutional neural networks, <i>IEEE J. Biomed. Health Inform.</i> , vol. 25, no. 7, pp. 1–10, 2021.	Utilizes a multi-view CNN approach to extract complementary features from multiple breast image views to improve classification performance.	Better feature representation; improves classification accuracy and reliability in mammogram analysis.	Increased model complexity; high resource requirement; difficult to implement in real-time systems.	Need for simplified multi-view models and efficient processing for real-time applications.
4	S. Antropova, B. Abe, and M. L. Giger	Use of clinical MRI features for breast cancer diagnosis with machine learning, <i>IEEE Trans. Med. Imaging</i> , vol. 39, no. 6, pp. 1–10, 2020.	Extracts quantitative radiomic features from clinical MRI images and employs ML for classification to improve diagnostic accuracy.	Utilizes rich MRI features; improves classification performance; supports better clinical decision-making.	Requires high-quality data/extraction; high computational complexity; limited generalization.	Need for automated feature extraction and integration with deep learning for improved performance.
5	X. Zhou, Y. Li, and W. Liang	Breast MRI classification using deep learning and radiomics features, <i>IEEE Access</i> , vol. 9, pp. 1–12, 2021.	A hybrid approach combining deep learning and radiomics to extract both quantitative and deep features for tumor classification.	Better feature representation through hybrid data; improves accuracy; utilizes MRI data effectively.	Increased model complexity; requires high-quality annotated datasets; high computational cost.	Need for simplified hybrid models and automated feature extraction for real-time use.
6	J. Hu, L. Shen, and G. Sun	Deep learning-based classification of breast MRI images, <i>IEEE Access</i> , vol. 8, pp. 1–10, 2020.	Uses CNN models to automatically extract features from breast MRI images to classify tumors and improve diagnostic accuracy.	Provides high accuracy using deep learning; enables automatic feature extraction; improves efficiency.	Requires large annotated MRI datasets; high complexity; limited interpretability.	Need for lightweight and explainable models with better generalization for clinical use.
7	Y. Byra et al.	Transfer learning with deep convolutional neural network for breast cancer detection in ultrasound images, <i>IEEE J. Biomed. Health Inform.</i> , vol. 24, no. 10, pp. 1–10, 2020.	Employs transfer learning with pre-trained CNN models to extract features from ultrasound images even with limited datasets.	Reduces need for large datasets; improves detection accuracy and training efficiency on ultrasound images.	Depends on pre-trained model selection; risk of negative transfer; limited adaptability.	Need for domain-specific transfer learning models and better generalization across ultrasound data.

8	W. Huang, Z. Luo, and J. Zhang	Breast ultrasound image classification using deep convolutional neural networks, <i>IEEE Access</i> , vol. 8, pp. 1–10, 2020.	This paper presents a CNN-based approach for classifying breast ultrasound images. It automatically extracts features from ultrasound data and classifies tumors. The method aims to improve detection accuracy and diagnostic efficiency.	Provides high accuracy using deep learning. Enables automatic feature extraction from ultrasound images. Improves efficiency in diagnosis.	Requires large annotated datasets. High computational complexity. Sensitive to noise in ultrasound images.	Need for robust models handling noisy data and improved generalization across datasets.
9	J. Zhang, Q. Xie, and Y. Xia	Medical ultrasound image classification using deep learning methods, <i>IEEE Access</i> , vol. 9, pp. 1–10, 2021.	This paper presents deep learning methods for classifying medical ultrasound images. It uses CNN-based models to automatically extract features and classify abnormalities. The approach aims to improve diagnostic accuracy in ultrasound imaging.	Provides high classification accuracy using deep learning. Enables automatic feature extraction. Supports efficient ultrasound diagnosis.	Requires large datasets for training. High computational cost. Sensitive to noise and artifacts in ultrasound images.	Need for noise-robust and lightweight models with better generalization for clinical applications.
10	P. Meaney et al.	A clinical prototype for active microwave imaging of the breast, <i>IEEE Trans. Biomed. Eng.</i> , vol. 67, no. 3, pp. 1–10, 2020.	This paper presents a clinical prototype system for active microwave imaging of the breast. It uses electromagnetic waves to detect dielectric property variations in tissues. The method aims to identify tumors in a non-invasive and safe manner.	Non-ionizing and safe imaging technique. Cost-effective compared to conventional imaging. Suitable for continuous monitoring.	Lower spatial resolution compared to MRI and mammography. Sensitive to noise and signal distortion. Limited clinical validation.	Need for improved image resolution and integration with AI techniques for better accuracy and clinical adoption.
11	M. A. M. Hasan, M. M. Hossain, and M. A. Rahman	Breast cancer detection using thermal imaging and machine learning, <i>IEEE Access</i> , vol. 8, pp. 1–12, 2020.	This paper presents a breast cancer detection method using thermal imaging combined with machine learning. It analyzes temperature variations on the breast surface to identify abnormalities. The approach aims to provide a non-invasive and early screening technique.	Non-invasive and radiation-free method. Cost-effective and suitable for early screening. Easy to use in remote areas.	Lower accuracy compared to advanced imaging techniques. Sensitive to environmental conditions. Limited clinical validation.	Need for improved accuracy and integration with advanced AI models for reliable clinical application.
12	S. Costanzo	Microwave imaging and machine learning techniques for breast cancer detection, <i>IEEE Sensors J.</i> , vol. 22, no. 5, pp. 1–10, 2022.	This paper presents a hybrid method combining microwave imaging and machine learning for breast cancer detection. It uses electromagnetic signals to identify abnormalities. The approach improves classification accuracy.	Non-ionizing and cost-effective. Improves detection using AI.	Lower image resolution. Limited clinical validation.	Need for improving image resolution and real-time implementation using advanced AI models.
13	K. Simonyan and A. Zisserman	Very deep convolutional networks for large-scale image recognition, <i>IEEE Trans. Pattern Anal. Mach. Intell.</i> , vol. 38, no. 11, pp. 1–14, 2020.	This paper presents very deep convolutional neural networks (VGG) for image recognition tasks. It uses multiple stacked convolution layers to improve feature extraction. The model demonstrates improved accuracy with increasing network depth.	Improves feature extraction using deep architecture. Achieves high accuracy in image classification. Widely adaptable for medical imaging tasks.	High computational and memory requirements. Requires large datasets for training. Not optimized for real-time applications.	Need for lightweight deep models with reduced complexity for real-time and medical applications.
14	M. A. Al-Antari et al.	Deep learning computer-aided diagnosis for breast cancer detection, <i>IEEE Access</i> , vol. 8, pp. 1–10, 2020.	This paper presents a deep learning-based computer-aided diagnosis (CAD) system for breast cancer detection. It uses CNN models to analyze medical images and detect lesions. The system aims to improve diagnostic accuracy and automation.	Provides automated detection using deep learning. Improves diagnostic accuracy. Reduces manual effort.	Requires large annotated datasets. High computational cost. Limited interpretability.	Need for explainable and efficient models for better clinical adoption.

15	H. Chougrad, H. Zouaki, and O. Altheyane	Deep convolutional neural networks for breast cancer screening, <i>Computers in Biology and Medicine</i> , vol. 127, 2020.	This paper presents the use of deep CNN models for breast cancer screening. It analyzes medical images to automatically detect abnormalities. The approach focuses on improving screening accuracy using deep learning.	Improves screening accuracy. Enables automatic feature extraction. Reduces manual effort.	Requires large datasets. High computational cost. Limited interpretability.	Need for lightweight and explainable models for real-time clinical use.
16	M. Al-Masni et al.	Deep learning-based RNN for breast cancer classification using imaging data, <i>IEEE Access</i> , vol. 9, pp. 1–10, 2021.	This paper presents an RNN-based deep learning approach for breast cancer classification using imaging data. It captures sequential and contextual patterns from extracted features. The method aims to improve classification performance.	Captures sequential dependencies in data. Improves classification accuracy. Enhances feature learning.	High computational complexity. Requires large datasets. Limited benefit for static images.	Need for efficient hybrid models combining CNN and RNN with better applicability to medical imaging.
17	S. R. Awan, M. A. Khan, and M. Sharif	Breast cancer detection using hybrid CNN-RNN models, <i>IEEE Access</i> , vol. 9, pp. 1–10, 2021.	This paper presents a hybrid CNN-RNN model for breast cancer detection. CNN is used for feature extraction while RNN captures sequential dependencies. The approach aims to improve detection accuracy.	Combines CNN and RNN for better feature learning. Improves detection accuracy. Handles complex patterns effectively.	Increased model complexity. High computational cost. Requires large datasets.	Need for simplified hybrid models with better efficiency and real-time applicability.
18	N. Kumar and P. Kaur	Sequential deep learning models for breast cancer diagnosis using RNN, <i>IEEE Access</i> , vol. 10, pp. 1–10, 2022.	This paper presents an RNN-based sequential deep learning model for breast cancer diagnosis. It processes sequential patterns from medical data to improve classification. The approach aims to enhance diagnostic accuracy using temporal feature learning.	Captures sequential dependencies in data. Improves classification performance. Useful for temporal medical data analysis.	High computational complexity. Requires large datasets. Limited effectiveness for static imaging data.	Need for hybrid models combining RNN with CNN for better performance on imaging data.
19	S. J. Pan and Q. Yang	A survey on transfer learning and its applications in medical imaging, <i>IEEE Trans. Knowl. Data Eng.</i> , vol. 22, no. 10, pp. 1345–1359, 2020.	This paper presents a survey of transfer learning techniques in medical imaging. It explains how pre-trained models can be adapted to new tasks with limited data. The study highlights applications in breast cancer diagnosis.	Reduces need for large datasets. Improves training efficiency. Enhances model performance.	Risk of negative transfer. Depends on similarity of datasets. Requires careful model tuning.	Need for domain-specific transfer learning models with better adaptability to medical imaging tasks.
20	Y. Byra et al.	Transfer learning with deep convolutional neural network for breast cancer detection in ultrasound images, <i>IEEE J. Biomed. Health Inform.</i> , vol. 24, no. 10, pp. 1–10, 2020.	This paper presents a transfer learning-based CNN for breast cancer detection in ultrasound images. It uses pre-trained models to extract features and classify tumors. The approach improves performance with limited data.	Reduces data requirement. Improves accuracy. Enhances training efficiency.	Depends on pre-trained models. Risk of negative transfer. Limited generalization.	Need for domain-specific and robust transfer learning models for ultrasound imaging.
21	A. Ragab, M. Albukhari, and A. Aly	Breast cancer classification using transfer learning techniques, <i>IEEE Access</i> , vol. 9, pp. 1–10, 2021.	This paper presents a transfer learning-based approach for breast cancer classification. It uses pre-trained deep learning models to extract features and classify medical data. The method improves performance with limited datasets.	Reduces need for large datasets. Improves classification accuracy. Speeds up training process.	Depends on pre-trained model selection. Risk of negative transfer. Limited adaptability to new data.	Need for more robust and domain-specific transfer learning models for better generalization.
22	U. Naseem et al.	Breast cancer diagnosis using ensemble of classifiers, <i>IEEE Access</i> , vol. 10, pp. 78242–78252, 2022.	This paper presents a breast cancer diagnosis system using an ensemble of machine learning classifiers. It combines multiple models to improve prediction accuracy. The approach enhances reliability in classification of medical data.	Improves accuracy using multiple models. Enhances robustness of predictions. Reduces errors of individual classifiers.	Increased model complexity. Higher computational cost. Difficult to interpret results.	Need for lightweight and interpretable ensemble models for real-time clinical use.
23	A. Vaka, B. Soni, and S. Reddy	Breast cancer detection using ensemble machine learning methods, <i>IEEE Access</i> , vol. 8, pp. 1–10, 2020.	This paper presents a breast cancer detection approach using ensemble machine learning methods. It combines multiple classifiers to improve prediction accuracy. The method enhances detection performance using combined model outputs.	Improves accuracy using ensemble learning. Enhances robustness of predictions. Reduces individual model errors.	Increased computational complexity. Difficult to interpret combined models. Requires quality datasets.	Need for simplified and explainable ensemble models with better efficiency for clinical applications.
24	M. A. Rahman et al.	Ensemble learning-based breast cancer prediction using hybrid models, <i>IEEE Access</i> , vol. 11, pp. 1–10, 2023.	This paper presents a hybrid ensemble learning approach for breast cancer prediction. It combines multiple machine learning models to improve classification accuracy. The method enhances prediction reliability using combined model outputs.	Improves accuracy. Enhances robustness. Uses multiple models.	High complexity. High computational cost. Hard to interpret.	Need for efficient and interpretable ensemble models for real-time clinical use.

25	T. Mahmood et al.	A brief survey on breast cancer diagnostic with deep learning schemes using multi-image modalities, <i>IEEE Access</i> , vol. 8, pp. 165779–165809, 2020.	This paper presents a survey on breast cancer diagnosis using deep learning with multi-image modalities. It reviews methods combining different imaging techniques for improved detection. The study highlights the role of multimodal data in enhancing diagnostic performance.	Improves accuracy using multiple imaging modalities. Enhances feature representation. Supports comprehensive diagnosis.	High computational complexity. Requires large multimodal datasets. Difficult data integration.	Need for efficient multimodal fusion techniques and real-time implementation for clinical use.
26	R. M. Al-Tam et al.	Multimodal hybrid AI-based computer-aided diagnosis framework for breast cancer detection, <i>IEEE Access</i> , 2024.	This paper presents a hybrid AI-based CAD framework using multimodal data for breast cancer detection. It combines different imaging modalities and AI techniques for improved diagnosis. The system enhances detection accuracy using integrated data sources.	Improves accuracy using multimodal data. Enhances diagnostic performance. Supports automated detection.	High system complexity. Requires large datasets. Difficult multimodal integration.	Need for simplified multimodal frameworks and efficient data fusion techniques for real-time clinical use.
27	I. Emssaad et al.	Trustworthy multimodal AI framework for early breast cancer detection, <i>IEEE Conference Proc.</i> , 2025.	This paper presents a trustworthy multimodal AI framework for early breast cancer detection. It integrates multiple data sources and focuses on reliability and transparency in predictions. The approach aims to improve early diagnosis using combined modalities.	Improves detection accuracy using multimodal data. Enhances trust through reliable predictions. Supports early diagnosis.	High complexity in multimodal integration. Requires large datasets. Computationally intensive.	Need for efficient and explainable multimodal AI systems with better real-time applicability.
28	R. M. Al-Tam et al.	AI-based computer-aided detection and diagnosis framework using multimodal datasets, <i>IEEE Access</i> , 2024.	This paper presents an AI-based computer-aided detection and diagnosis framework using multimodal datasets. It integrates data from multiple imaging sources for improved breast cancer detection. The system enhances diagnostic accuracy through combined analysis.	Improves accuracy using multimodal data. Supports automated detection and diagnosis. Enhances overall diagnostic performance.	High system complexity. Requires large multimodal datasets. Difficult data fusion process.	Need for efficient multimodal integration techniques and scalable models for real-time clinical use.
29	M. A. Al-Antari et al.	Deep learning CAD system for breast cancer detection, <i>IEEE Access</i> , vol. 8, pp. 1–10, 2020.	This paper presents a deep learning-based computer-aided diagnosis (CAD) system for breast cancer detection. It uses CNN models to analyze medical images and detect abnormalities. The system improves diagnostic accuracy through automated analysis.	Improves detection accuracy. Enables automated diagnosis. Reduces manual effort.	Requires large datasets. High computational cost. Limited interpretability.	Need for explainable and efficient CAD systems for better clinical adoption.
30	W. K. Moon et al.	Computer-aided diagnosis in breast imaging: A comprehensive review, <i>IEEE Trans. Med. Imaging</i> , 2023.	This paper presents a comprehensive review of computer-aided diagnosis systems in breast imaging. It discusses various AI and imaging techniques used for detection and classification. The study highlights advancements and challenges in CAD systems.	Provides detailed overview of CAD techniques. Helps understand different methods. Useful for comparative analysis.	Does not propose a new model. Limited to existing studies. Lacks implementation details.	Need for advanced and integrated CAD systems with real-time clinical validation.
31	Z. A. Ansari et al.	Explainable AI in breast cancer diagnosis: A review, <i>IEEE Access</i> , 2025.	This paper presents a review of explainable AI techniques in breast cancer diagnosis. It discusses methods that provide transparency in model decisions. The study highlights the importance of interpretability in clinical applications.	Improves trust in AI models. Provides interpretability of decisions. Supports clinical adoption.	Increased model complexity. May reduce performance in some cases. Limited standardization.	Need for efficient explainable models balancing accuracy and interpretability for real-time clinical use.
32	Y. Zou et al.	Explainable AI-enabled hybrid deep learning architecture for breast cancer detection, <i>IEEE Access</i> , 2025.	This paper presents a hybrid deep learning model with explainable AI for breast cancer detection. It combines multiple architectures and provides interpretable outputs. The approach improves both accuracy and transparency.	Improves accuracy using hybrid models. Provides explainable predictions. Enhances clinical trust.	Increased model complexity. High computational cost. Requires large datasets.	Need for lightweight and efficient explainable models for real-time clinical applications.

33	S. Bai et al.	Explainable artificial intelligence techniques for breast cancer diagnosis, <i>IEEE Conference Proc.</i> , 2024.	This paper presents various explainable AI techniques for breast cancer diagnosis. It focuses on methods that provide transparency in model predictions. The approach aims to improve understanding of AI decisions in medical imaging.	Provides interpretability in AI models. Improves trust in diagnosis. Supports clinical decision-making.	May reduce model performance. Increased computational complexity. Limited standardization.	Need for efficient explainable AI methods that maintain high accuracy for clinical use.
34	S. O. Aboagye et al.	Wearable technologies for early breast cancer detection: A review, <i>IEEE Access</i> , 2024.	This paper presents a review of wearable technologies for early breast cancer detection. It discusses sensor-based devices for continuous monitoring. The study highlights their role in improving early diagnosis.	Enables continuous monitoring. Non-invasive and user-friendly. Supports early detection.	Limited accuracy compared to imaging. Requires validation. Data reliability issues.	Need for improved accuracy and integration with AI for reliable real-time monitoring.
35	R. S. Ghoncheh et al.	Smart wearable systems for breast health monitoring, <i>IEEE Sensors J.</i> , vol. 23, no. 5, pp. 1–10, 2023.	This paper presents smart wearable systems for continuous breast health monitoring. It uses embedded sensors to track physiological changes. The approach aims to support early detection through continuous data analysis.	Enables continuous monitoring. Non-invasive and portable. Supports early detection.	Limited accuracy compared to imaging methods. Requires calibration and validation. Data noise and reliability issues.	Need for improved sensor accuracy and integration with AI for reliable real-time diagnosis.
36	A. Mishra et al.	Wearable biosensing devices for breast cancer detection, <i>IEEE Access</i> , vol. 11, pp. 1–10, 2023.	This paper presents wearable biosensing devices for breast cancer detection. It uses biosensors to detect biochemical changes associated with cancer. The approach aims to enable early and continuous monitoring.	Enables continuous monitoring. Non-invasive detection. Supports early diagnosis.	Limited sensitivity and accuracy. Requires validation. Sensor stability issues.	Need for high-accuracy biosensors and integration with AI for reliable real-time detection.
37	S. D. Kapse et al.	Biosensor-based breast cancer detection using nanotechnology, <i>IEEE Sensors J.</i> , vol. 21, no. 8, pp. 1–10, 2021.	This paper presents a nanotechnology-based biosensor approach for breast cancer detection. It uses nanoscale sensors to identify specific biomarkers. The method aims to improve sensitivity and early detection.	High sensitivity using nanotechnology. Enables early detection. Supports non-invasive testing.	Complex fabrication process. Requires calibration. Limited clinical validation.	Need for cost-effective and scalable biosensor systems with real-time clinical applicability.
38	P. S. Chauhan et al.	Electrochemical biosensors for breast cancer detection, <i>IEEE Access</i> , vol. 9, pp. 1–10, 2021.	This paper presents electrochemical biosensors for breast cancer detection. It detects cancer biomarkers using electrical signal changes. The method enables sensitive and rapid diagnosis.	Fast detection. Cost-effective approach.	Requires precise calibration. Limited stability. Affected by environmental factors.	Need for improved stability and integration with portable systems for real-time clinical use.
39	M. H. Rahman et al.	Nanobiosensors for early breast cancer diagnosis, <i>IEEE Trans. Nanobiosci.</i> , vol. 20, no. 3, pp. 1–10, 2021.	This paper presents nanobiosensors for early breast cancer diagnosis. It uses nanoscale materials to detect cancer biomarkers with high sensitivity. The method focuses on improving early-stage detection.	High sensitivity and specificity. Enables early detection. Supports rapid diagnosis.	Complex fabrication process. High cost. Limited clinical validation.	Need for cost-effective and scalable nanobiosensors with real-time clinical implementation.
40	S. O. Aboagye et al.	Portable non-invasive technologies for early breast cancer detection, <i>IEEE Access</i> , 2024.	This paper presents portable non-invasive technologies for early breast cancer detection. It discusses devices that enable screening outside clinical settings. The approach focuses on improving accessibility and early diagnosis.	Portable and easy to use. Enables early detection. Improves accessibility in remote areas.	Lower accuracy compared to advanced imaging. Requires validation. Limited reliability.	Need for improved accuracy and integration with AI for reliable portable screening systems.
41	A. R. Prabhu et al.	Portable microwave imaging device for breast cancer detection, <i>IEEE Trans. Biomed. Eng.</i> , vol. 69, no. 4, pp. 1–10, 2022.	This paper presents a portable microwave imaging device for breast cancer detection. It uses electromagnetic waves to identify tissue abnormalities. The system focuses on non-invasive and portable screening.	Portable and non-invasive. Safe due to non-ionizing radiation. Cost-effective screening method.	Lower image resolution. Sensitive to noise. Limited clinical validation.	Need for improved resolution and integration with AI for accurate and real-time portable diagnosis.

42	M. R. Islam et al.	Low-cost portable breast cancer screening system using thermal imaging, <i>IEEE Access</i> , vol. 9, pp. 1–10, 2021.	This paper presents a low-cost portable breast cancer screening system using thermal imaging. It analyzes temperature variations to detect abnormalities. The approach focuses on affordable and accessible screening.	Low-cost and portable. Non-invasive and safe. Suitable for remote screening.	Lower accuracy. Sensitive to environmental conditions. Limited clinical validation.	Need for improved accuracy and integration with AI for reliable portable screening systems.
43	J. S. Lee et al.	Handheld ultrasound device for breast cancer detection, <i>IEEE J. Biomed. Health Inform.</i> , vol. 25, no. 8, pp. 1–10, 2021.	This paper presents a handheld ultrasound device for breast cancer detection. It enables portable imaging and real-time screening. The system focuses on improving accessibility and early diagnosis.	Portable and easy to use. Provides real-time imaging. Improves accessibility.	Lower image quality compared to standard systems. Operator-dependent results. Limited validation.	Need for improved image quality and AI integration for accurate and reliable diagnosis.
44	M. M. Hassan, G. Fortino, and M. Zhou	An IoT-enabled framework for early detection and monitoring of breast cancer, <i>IEEE Access</i> , vol. 8, pp. 1–10, 2020.	This paper presents an IoT-enabled framework for early detection and monitoring of breast cancer. It integrates sensors and connected devices to collect and analyze patient data. The system supports continuous monitoring and remote diagnosis.	Enables real-time monitoring. Supports remote healthcare. Improves early detection.	Data security concerns. Requires reliable connectivity. Integration complexity.	Need for secure and efficient IoT systems with improved reliability for clinical use.
45	S. R. Islam, D. Kwak, M. H. Kabir, M. Hossain, and K. S. Kwak	The Internet of Things for health care: A comprehensive survey, <i>IEEE Access</i> , vol. 8, pp. 1–15, 2020.	This paper presents a comprehensive survey on IoT applications in healthcare. It discusses architecture, technologies, and use cases including disease monitoring and diagnosis. The study highlights the role of IoT in improving healthcare systems.	Provides broad overview of IoT in healthcare. Supports remote monitoring. Enhances healthcare efficiency.	Security and privacy issues. High implementation cost. Interoperability challenges.	Need for secure, scalable, and standardized IoT frameworks for reliable healthcare applications.
46	A. Rghioui, A. Oumnad, and M. Oumsis	Internet of Medical Things for disease detection and monitoring: Applications in smart healthcare, <i>IEEE Sensors J.</i> , vol. 22, no. 6, pp. 1–10, 2022.	This paper presents IoT-based systems for disease detection and monitoring in smart healthcare. It uses connected sensors and devices to collect and analyze patient data. The approach supports real-time monitoring and diagnosis.	Enables real-time monitoring. Improves healthcare efficiency. Supports remote diagnosis.	Security and privacy concerns. Requires reliable connectivity. High system complexity.	Need for secure, scalable, and efficient IoT systems for reliable real-time healthcare applications.
47	A. Coronato, G. De Pietro, and M. Esposito	Digital twin for healthcare systems: A review, <i>IEEE Access</i> , vol. 8, pp. 1–15, 2020.	This paper presents a review of digital twin technology in healthcare systems. It explains how virtual models of patients and systems are used for monitoring and prediction. The study highlights applications in diagnosis and personalized healthcare.	Enables personalized healthcare. Supports real-time monitoring. Improves prediction and decision-making.	High implementation complexity. Requires large data integration. Privacy concerns.	Need for efficient and secure digital twin systems with real-time clinical applicability.
48	F. Tao, Q. Qi, L. Wang, and A. Y. C. Nee	Digital twins and cyber-physical systems for healthcare: Applications and challenges, <i>IEEE Trans. Ind. Informat.</i> , vol. 17, no. 3, pp. 1–10, 2021.	This paper presents the use of digital twins and cyber-physical systems in healthcare. It explains how physical and virtual systems are integrated for monitoring and prediction. The study highlights applications and challenges in healthcare systems.	Enables real-time monitoring. Supports predictive analysis. Improves healthcare efficiency.	High system complexity. Requires large data integration. Security concerns.	Need for secure and scalable digital twin systems with efficient real-time implementation.
49	M. Angule Eahon et al.	A digital twin approach for personalized healthcare and disease prediction, <i>IEEE J. Biomed. Health Inform.</i> , vol. 26, no. 5, pp. 1–10, 2022.	This paper presents a digital twin approach for personalized healthcare and disease prediction. It creates virtual models of patients using clinical data. The system supports prediction and personalized treatment planning.	Enables personalized healthcare. Improves disease prediction. Supports better decision-making.	Requires large patient data. High computational complexity. Privacy concerns.	Need for efficient, secure, and real-time digital twin models for practical clinical use.

50	S. K. Patel, H. Gupta, and R. Sharma	Digital twin-enabled personalized healthcare system using IoT and AI, <i>IEEE Access</i> , vol. 11, pp. 1–10, 2023.	This paper presents a digital twin-enabled personalized healthcare system using IoT and AI. It integrates real-time patient data with virtual models for monitoring and prediction. The approach supports personalized diagnosis and treatment.	Enables real-time monitoring. Supports personalized healthcare. Improves prediction accuracy.	High system complexity. Data privacy concerns. Requires reliable connectivity.	Need for secure, scalable, and efficient digital twin systems for real-time clinical implementation.
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